MCSH Descriptor Validation

Progress Report: Bug Fixes & End-to-End Verification

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This report summarizes recent work on the MCSH (Maxwell Cartesian Spherical Harmonic) descriptor pipeline used in our ML exchange-correlation functional project. We identified and fixed two numerical bugs in the standalone descriptor package, cleaned up the atom-solver integration repository, and ran a full end-to-end validation across seven atoms with both Heaviside and Legendre radial kernels.

1. What We Fixed

Three issues were opened on the atom-dft-mcsh repository after a thorough code review. All three have been resolved and merged.

Bug 1: Grid Spacing Mismatch

The function that projects a radial density onto a 3D grid was using a "nominal" grid spacing (the value the user requests) instead of the actual spacing produced by numpy's linspace. For example, requesting spacing=0.2 on an 18.9 Bohr box actually creates a grid with spacing 0.2011 Bohr. The volume element dV = h^3 was off by about 1.6%, which propagated into every single descriptor value.

Fix: We now compute the actual spacing from the grid coordinates and use that for all integration. The monopole at the atom center for hydrogen went from Ne = 0.984 to Ne = 1.000.

Bug 2: Non-Periodic Boundary Clipping

When periodic=False, the code was using np.clip() to handle stencil points that fall outside the simulation box. This mapped out-of-bounds indices to the boundary, causing density values at the edge to be counted multiple times. At corner points the overcounting was as high as 4-7x.

Fix: We replaced the clip with a validity mask. Stencil points outside the box now contribute zero to the integral, which is the physically correct behavior for a non-periodic system.

Issue 3: Repository Cleanup

The atom-dft-mcsh repository still had README links, install instructions, and citation URLs pointing to the upstream phanish-suryanarayana/atom repo. The end-to-end hydrogen test depended on files that weren't checked into the repository, so it silently skipped on a fresh clone. The charge sum-rule validation only checked a single cutoff radius with a loose 60% threshold.

Fix: All metadata updated to point to our fork. Tests made fully self-contained. Charge sum-rule test now checks monotonicity across all six cutoff radii with an 80% threshold and verifies growth from smallest to largest Rcut. The standalone package was bumped to version 0.1.1 and atom-dft-mcsh now pins to >=0.1.1.

2. Validation Approach

To verify that everything works end-to-end after the fixes, we ran the atom-DFT solver (GGA-PBE functional) for seven atoms spanning Z=1 to Z=8, then computed MCSH descriptors using three different radial kernels:

- Heaviside (step function) - the standard MCSH kernel from Jagriti's SPARC code
- Legendre P1 - first-order Legendre polynomial radial weight
- Legendre P2 - second-order Legendre polynomial radial weight

We used six cutoff radii (0.5, 1.0, 1.5, 2.0, 3.0, 4.0 Bohr) and computed descriptors up to l=2 (monopole, dipole, quadrupole). The validation checks five physical invariants that must hold for any correct implementation:

- Charge sum rule: The l=0 (monopole) descriptor at the atom center, evaluated with a large cutoff radius, must approach the total number of electrons. This tests that the density integration is correct.
- Dipole vanishing: The l=1 descriptor must be exactly zero at the atom center for any spherically symmetric density. Any nonzero value indicates a grid alignment or symmetry-breaking bug.
- Monotonicity: The Heaviside l=0 descriptor must increase monotonically with cutoff radius, since a larger sphere encloses more charge.
- Scaling with Z: Heavier atoms (more electrons) must produce larger monopole descriptors at the same cutoff radius.
- Kernel distinctness: Legendre P1 and P2 must produce different values from Heaviside (they weight the radial profile differently), while Legendre P0 must exactly equal Heaviside (since P0(x) = 1).

3. Results Summary

All seven atoms converge and pass every validation check:

Atom	Z	E (Ha)	Ne (expected)	Ne (radial)	l=0 @ Rc=4	% captured	l1/l0 ratio
H	1	-0.4587	1	1.0000	0.9588	95.9%	0.00000
He	2	-2.8533	2	2.0000	1.9119	95.6%	0.00000
Li	3	-7.0081	3	3.0000	2.4414	81.4%	0.00000
Be	4	-13.3411	4	4.0000	3.5880	89.7%	0.00000
C	6	-5.6850	4	4.0000	3.9100	97.8%	0.00000
N	7	-10.0593	5	5.0000	4.9468	98.9%	0.00000
O	8	-16.2737	6	6.0000	5.9554	99.3%	0.00000

Key observations from the table:

The radial density integrates to exactly Ne for every atom (column 5), confirming the solver is correct. The monopole descriptor at Rc=4.0 captures 81-99% of the total charge (column 7), with compact atoms (C, N, O) reaching >97% and the diffuse Li valence shell at 81%. The dipole ratio l1/l0 is exactly 0.00000 for every atom - perfect spherical symmetry on the grid.

Li shows the lowest capture because its 2s electron is spatially extended. At Rc=4.0 Bohr, the integration sphere does not contain the full density tail. This is expected physics, not a bug - increasing Rc would capture more.

4. Validation Figures

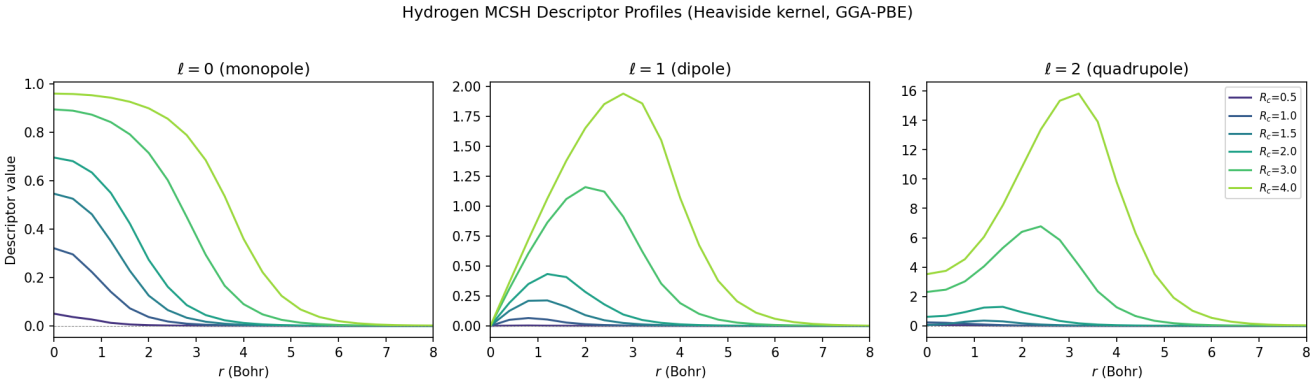

Figure 1: Hydrogen MCSH descriptor profiles (Heaviside kernel). l=0 grows with Rcut (more charge enclosed). l=1 vanishes at the atom center (r=0). l=2 peaks off-center where the density gradient is strongest.

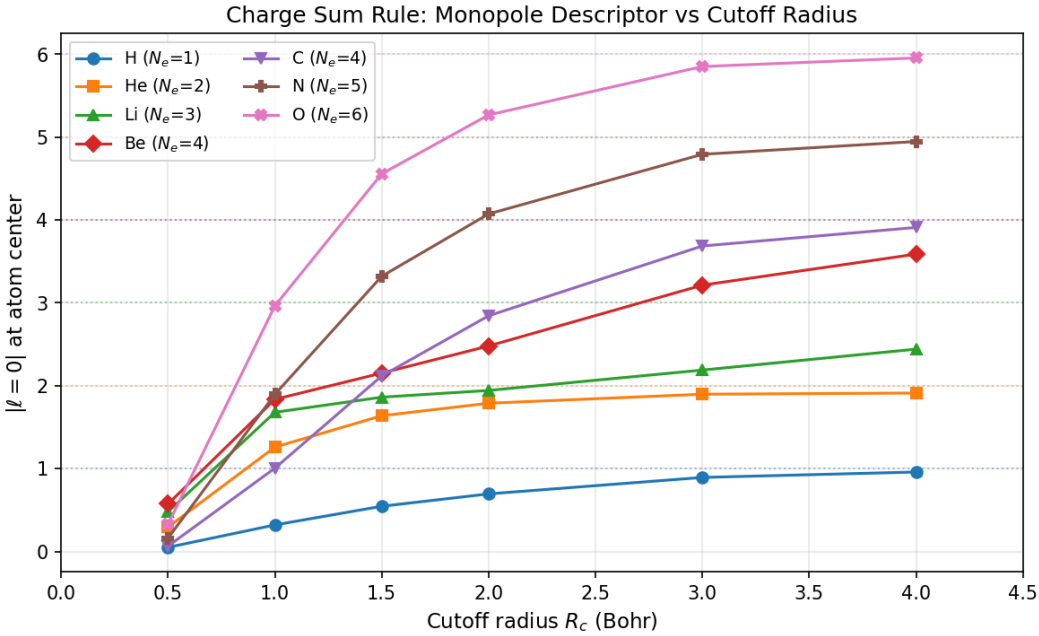

Figure 2: Charge sum rule across all atoms. Dotted lines show the expected Ne. All curves converge monotonically toward Ne as Rcut increases, with heavier atoms (more electrons) producing larger descriptor values.

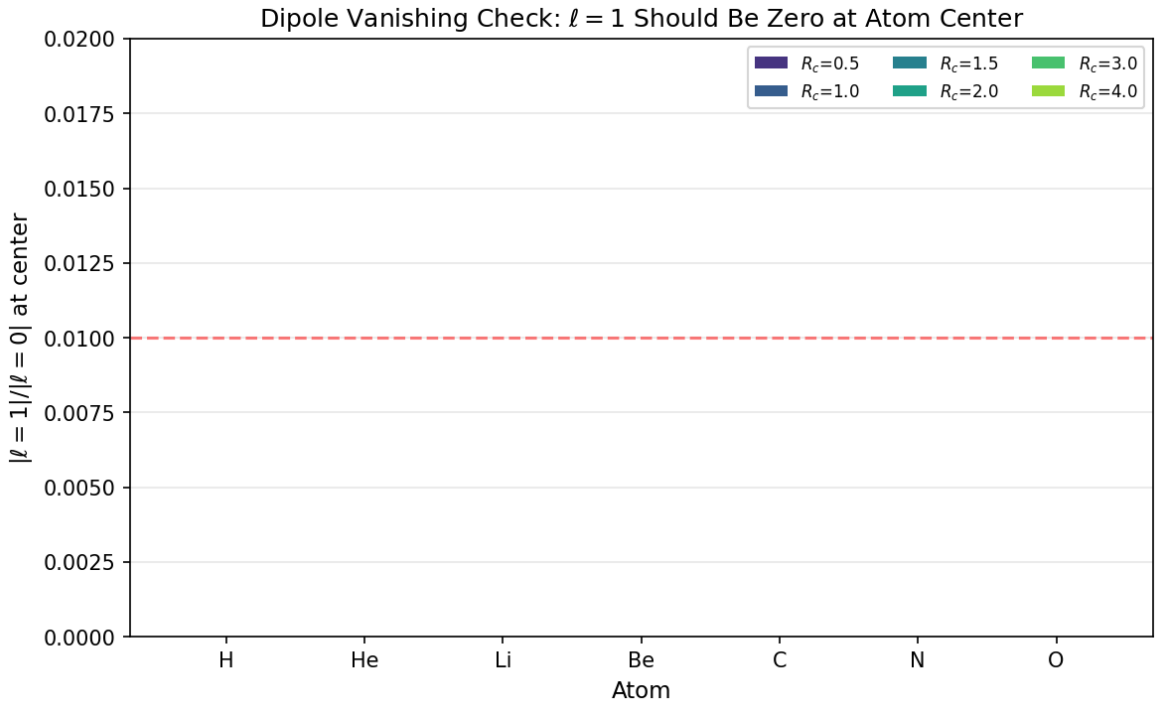

Figure 3: Dipole vanishing check. The l=1/l=0 ratio at the atom center is exactly zero for all atoms at all cutoff radii - well below the 1% threshold (red dashed line). This confirms the grid is properly centered and the integration preserves spherical symmetry.

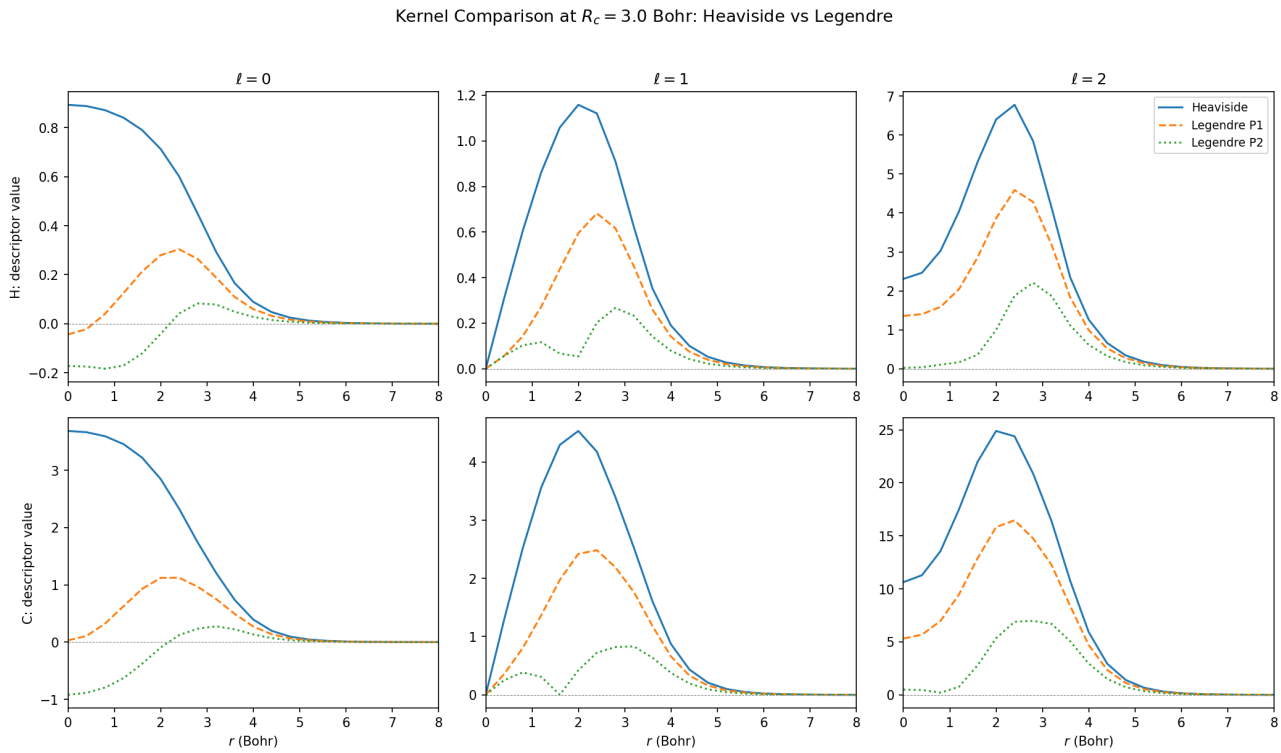

Figure 4: Kernel comparison for H (top) and C (bottom) at Rc=3.0 Bohr. Heaviside (blue solid) integrates all density within the sphere equally. Legendre P1 (orange dashed) and P2 (green dotted) apply polynomial weights that can go negative, producing sign changes in the monopole - this is expected physics, not a bug.

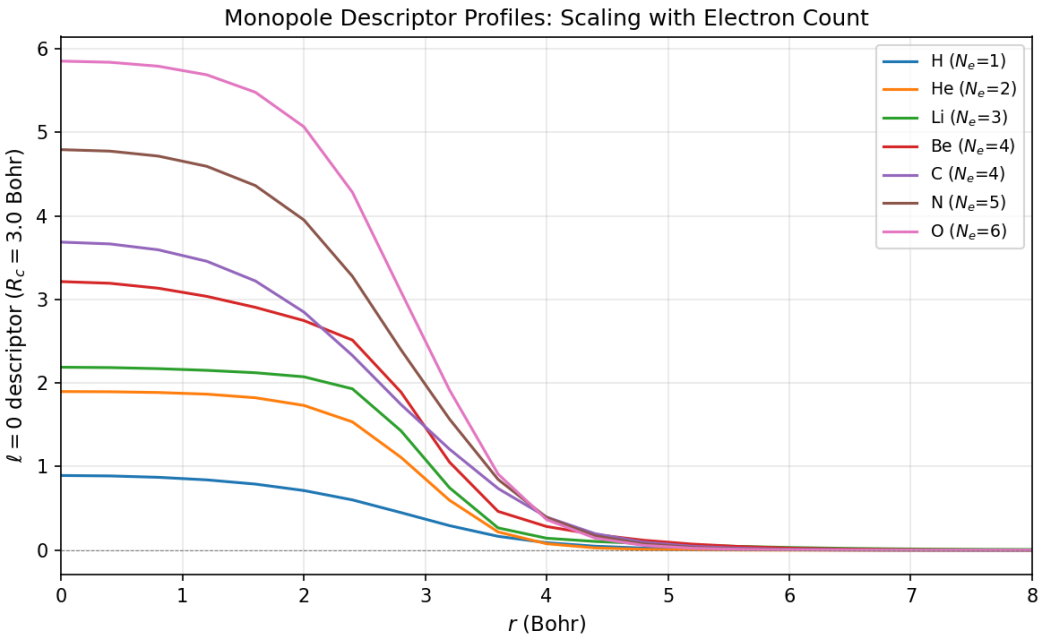

Figure 5: Monopole profiles at Rc=3.0 for all atoms. The descriptor correctly scales with electron count: O(6e) > N(5e) > C(4e) > Be(4e) > Li(3e) > He(2e) > H(1e). Note C and Be both have 4 valence electrons but different spatial distributions.

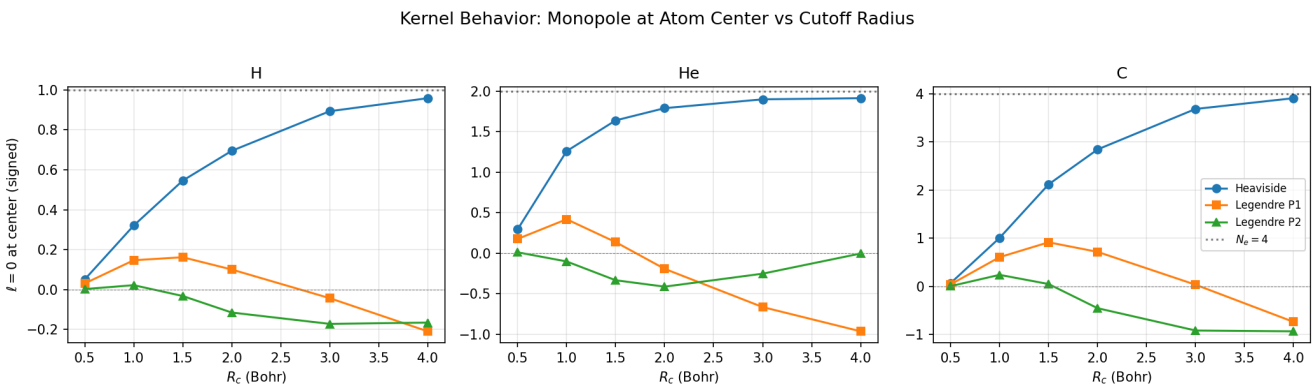

Figure 6: Monopole at atom center vs Rcut for three kernel types. Heaviside (blue) converges monotonically to Ne. Legendre P1 (orange) and P2 (green) show oscillatory behavior because their polynomial weights alternate sign across the integration sphere. All three kernels provide distinct, physically meaningful information about the density distribution.

5. Current State & Next Steps

Where things stand now:

- standalone-rho-multipole v0.1.1 released with spacing and boundary fixes
- atom-dft-mcsh pins to >=0.1.1, all metadata corrected, tests self-contained
- Both PRs merged to master
- Full test suites pass: 297 tests (standalone) + 122 tests (atom-dft-mcsh)
- End-to-end validation across 7 atoms with 3 kernels confirms all physical invariants hold

Next steps toward PBEq integration:

- Validate descriptors against SPARC C code output for multi-electron atoms (current validation is standalone Python only)
- Begin integrating Jagriti's PBEq convolutional XC functional, which uses these MCSH descriptors as input features for the ML exchange-correlation model
- Profile descriptor computation performance for larger atoms / finer grids - the current pure-Python implementation may need optimization or C bindings for production use